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PREMIUM NOTES

SSC — Quantitative Aptitude Formula Notes

SSC Test Series — StuDy

High-yield facts, formulas and trap-radar — made for the real exam hall.

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SSC — Quantitative Aptitude Formula Notes

Percentage & Profit–Loss

- x% increase then x% decrease ⇒ net LOSS of $x^2/100$ %. (10% up, 10% down = 1% loss.)
- Same selling price, one at +x% and one at −x% ⇒ ALWAYS loss = $x^2/100$ %.
- Successive discounts a% and b% = single discount of a + b − ab/100 %.
- Marked price: SP = MP × (1 − d/100); Profit% is always on CP, Discount% always on MP.
- If CP of x articles = SP of y articles (x>y), profit% = (x−y)/y × 100.

Simple & Compound Interest

- SI = PRT/100. CI − SI for 2 years = $P(r/100)^2$. For 3 years = $P(r/100)^2 \times (3 + r/100)$.
- Amount under CI: A = P(1 + r/100)^t. Half-yearly: rate r/2, time 2t. Quarterly: r/4, 4t.
- CI for the 3rd year alone = CI(3 yrs) − CI(2 yrs) amounts' difference — compute amounts, don't guess.
- Installments: $P = x/(1+r/100) + x/(1+r/100)^2 + \ldots$ for equal annual installment x.
- Fractional years (e.g. 2⅖ yrs): compound for full years, then SI for the fraction on that amount.

Time, Work & Pipes

- Use LCM method: total work = LCM of days; efficiency = work/days. Combined days = LCM / (sum of efficiencies).
- A leaves after k days: remaining work = total − (A's eff × k); time for B = remaining / B's eff.
- Man-days rule: $M_1D_1/W_1 = M_2D_2/W_2$. Wages divide in the ratio of EFFICIENCY (work per day), not days.
- A pipe that empties has NEGATIVE efficiency. Net rate decides fill/empty.

Speed, Trains & Boats

- km/h → m/s: × 5/18. m/s → km/h: × 18/5. (The most-missed conversion in SSC.)
- Train crossing a pole = its own length; crossing a platform = train length + platform length.
- Two trains towards each other: relative speed = SUM; same direction: DIFFERENCE.
- Boat: downstream = b + s, upstream = b − s. Boat speed = (down + up)/2; stream = (down − up)/2.
- Average speed for equal distances = 2xy/(x+y) (harmonic mean) — never (x+y)/2.

Ratio, Average, Mixtures

- Ratio stitching A:B and B:C — first make B EQUAL (LCM) in both, then combine. Skipping this is the classic trap.
- Average: when a person joins/leaves, use TOTAL sums, not averages. New avg = (old total ± change) / new count.
- Alligation: (cheaper qty)/(dearer qty) = (dearer price − mean)/(mean − cheaper price) — cross-difference.
- Replacement mixture: after n operations, quantity left = $x(1 - y/x)^n$.

Geometry & Mensuration — Fixed Stars

STUDY

- Inradius of triangle $r = \Delta/s$; Circumradius $R = abc/4\Delta$. For equilateral side a: $r = a/(2\sqrt{3})$, $R = a/\sqrt{3}$ ($R = 2r$).
- Angles: exterior angle = sum of the two remote interiors. Angle in a semicircle is a right angle.
- Cylinder TSA $= 2\pi r(r+h)$, CSA $= 2\pi rh$; Cone CSA $= \pi rl$, $l = \sqrt{(r^2+h^2)}$; Sphere SA $= 4\pi r^2$, $V = 4/3\pi r^3$.
- Hemisphere TSA $= 3\pi r^2$ (includes base), CSA $= 2\pi r^2$.
- Similar triangles: ratio of areas = square of ratio of sides.

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